

Analysis of the characteristics of the Ingeo™ 3D850 polymer in the context of its use in visual inspections in the Triga IPR-R1 nuclear reactor

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1. INTRODUCTION

Reliable underwater visual inspections at the TRIGA IPR-R1 require auxiliary parts like jigs, spacers, sensor mounts that can be rapidly fabricated and dimensionally stable near ionizing radiation fields. Ingeo™ Biopolymer 3D850 (PLA-HT) is attractive for such fixtures because it prints well and offers higher heat resistance and dimensional stability than conventional PLA (Ingeo™ 4043D). Available data shows 3D850 has a higher peak melting temperature ($\approx 165\text{--}180\text{ }^{\circ}\text{C}$ vs. $145\text{--}160\text{ }^{\circ}\text{C}$) and markedly higher heat-distortion temperature ($\approx 80\text{--}90\text{ }^{\circ}\text{C}$ vs. $55\text{ }^{\circ}\text{C}$), despite a somewhat lower nominal tensile strength than 4043D ($\approx 51\text{ MPa}$ vs. 60 MPa).

Polymers can undergo chain scission, oxidative degradation, and/or crosslinking under gamma radiation, those mechanisms can alter tensile and thermal behavior. Application near reactor fields demands dose-response evidence rather than catalog properties.

2. MATERIALS AND METHODS

Dose rates near the vicinity of a nuclear fuel element were measured, simulating the conditions under which printed components would be subjected to gamma radiation. Measures of the dose rate inside the central tube of the Triga IPR-R1 were also taken.

According to internal documentation from CDTN (2025), measurements were carried out using a Farmer 2502/3 dosimeter, using a $0,6\text{ cm}^3$ sensor. The reactor was in criticality 3 days prior to the measuring. The resulting data is presented in Tables 2 and 3.

Table 2: Dose rate inside the reactor's central tube

Time (h)	Exposure (R)	Exposure Rate	Dose Rate (Gy/h)
0,035	727	20937,6	183,6
0,034	704	20773,8	182,1
0,034	702,5	20729,5	181,8
0,035	719	20542,9	180,1

Table 3: Dose rate in the fuel element vicinity

	Time (h)	Exposure (R)	Exposure rate	Dose rate (Gy/h)
Fuel element	0,018	60,5	3403,1	29,8
Background	0,049	10	203,4	1,8
Liquid dose rate			3200	28

According to internal documentation from CDTN, measurements were carried out to determine gamma dose rates at the surface of the reactor well with one of the fuel elements at various depths. Fuel element was vertically repositioned in increments, ranging from a depth of 2,7 meters up to 50 cm.

A total of 50 samples were fabricated using PLA-HT filament. All samples were printed in accordance with the dimensional specifications defined for "Type V" samples, as outlined on pages 3 and 4 of ASTM D638 – 14. Of these samples, 45 were subjected to irradiation, grouped in sets of five, forming nine irradiated groups and one control group.

The irradiated groups were exposed to gamma radiation doses of 7 kGy, 14 kGy, 21 kGy, 28 kGy, 35 kGy, 50 kGy, 93 kGy, 130 kGy, and 175 kGy at the Gamma Irradiation Laboratory (LIG), located within the CDTN facilities, which operates a Category II panoramic multipurpose irradiator.

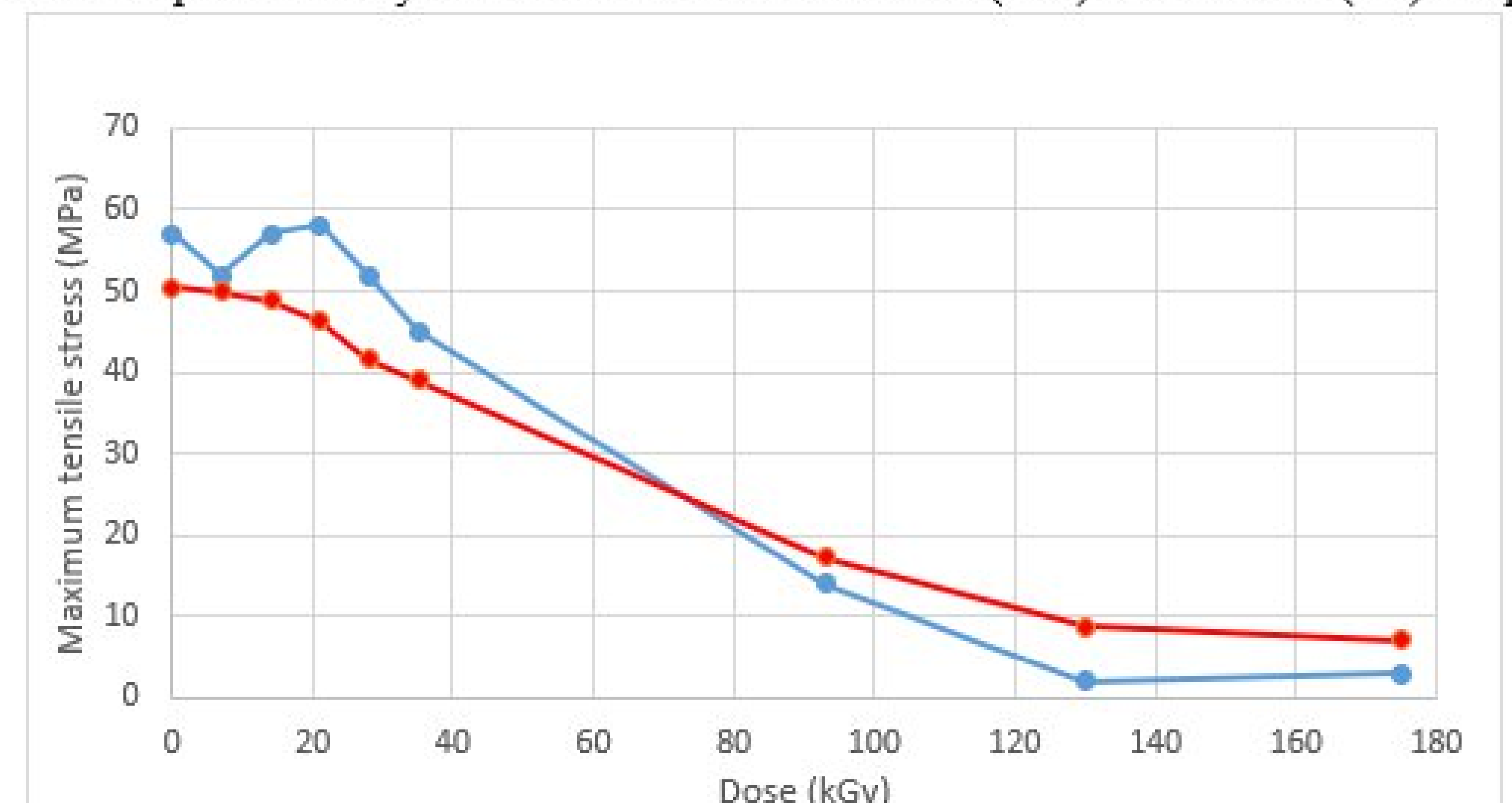
Following irradiation, tensile tests were conducted at the Mechanical Testing Laboratory of CDTN.

3. RESULTS

From the analysis of the resulting data, a pronounced embrittlement effect is evident in the material when exposed to gamma radiation, particularly at higher dose levels.

A comparative analysis was also performed using data reported by Alsabbagh (2020), represented in Figure 2. These results indicate that the Ingeo™ 4043D material exhibits superior tensile strength performance up to approximately 70 kGy. Beyond this dose range, Ingeo™ 3D850 demonstrates slightly enhanced mechanical resistance under tensile loading.

Figure 2: A comparative analysis was conducted between PLA (blue) and PLA-HT (red) samples.



Based on the previously presented data, a comparative assessment between the two materials can be made, taking into account the onset threshold at which each material begins to exhibit a significant reduction in tensile strength, as well as their suitability for applications within the operational environment of the Triga IPR-R1 reactor. These considerations are summarized in Table 5.

Table 5: Time estimates for PLA and PLA-HT under the Triga working conditions

	Significant embrittlement threshold (kGy)	Estimated Time on fuel element vicinity (h)	Estimated time on reactor central tube (h)	Estimated time 50 cm outside the reactor with fuel at 50 cm underwater (h)
PLA	28	1000	153	16887
PLA-HT	21	750	115	12665

4. CONCLUSIONS

PLA-HT exhibits dose-dependent embrittlement with a practical threshold near 21 kGy; it can be used for limited-duration inspection fixtures if exposure is managed by location and time. For lower-dose regimes ($< \sim 70\text{ kGy}$), standard PLA may retain higher tensile strength; however, PLA-HT offers advantages in dimensional stability/ductility and becomes comparatively favorable at higher doses. Incorporating material-specific radiation response and local dose-rate mapping into design is essential to assure mechanical reliability of 3D-printed components used around TRIGA IPR-R1 fuel inspections.

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